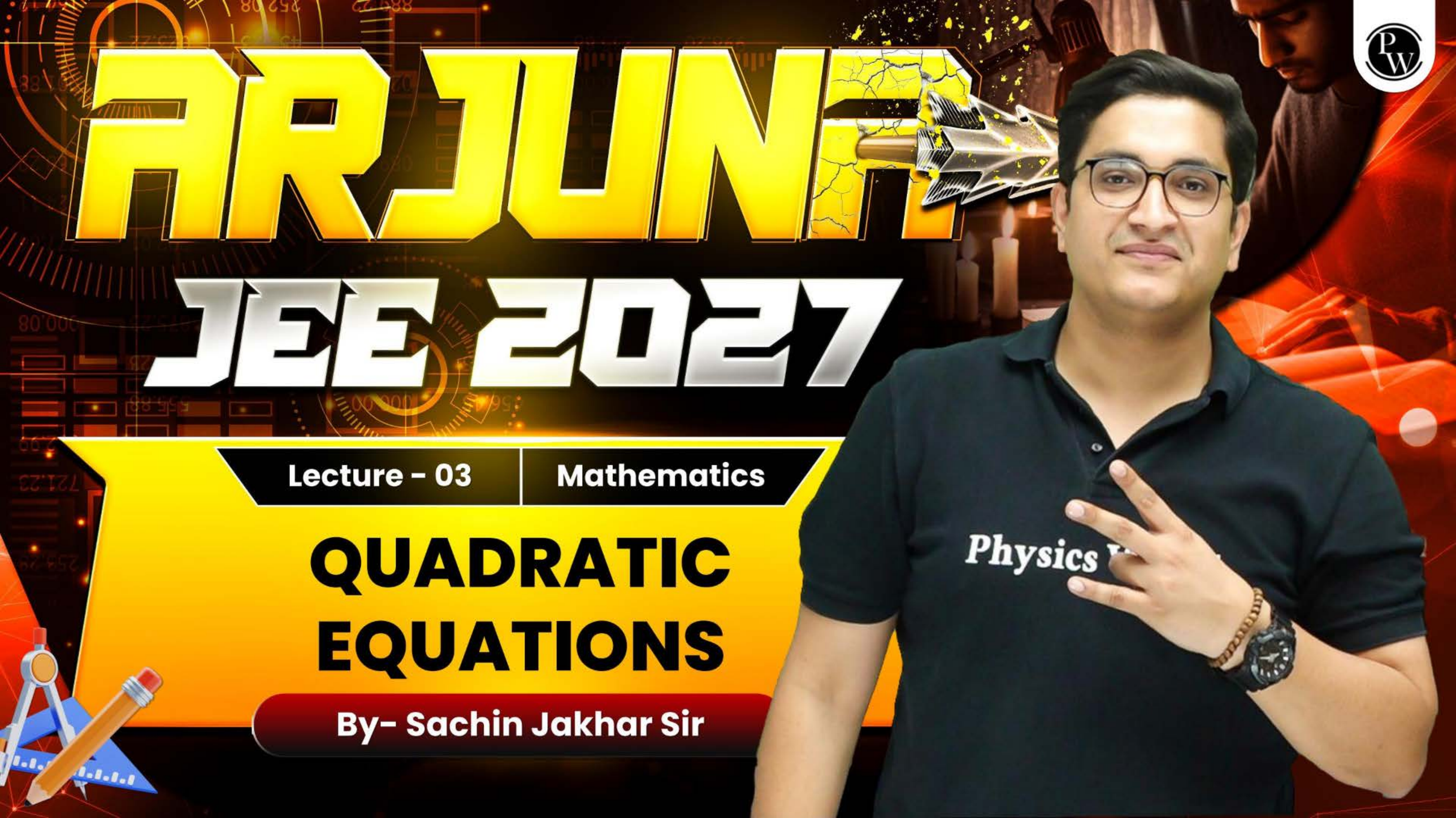
ARJUNA JEE 2027

Lecture - 03 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A Symmetric Functions of Roots

B Newton's Formula

C Question Practice

D Note: Basic Mathematics DIBY PDF is uploaded





LAST CLASS RECAP

Answer the following Questions:

1. $\frac{(x+b)(x+c)}{(b-a)(c-a)} + \frac{(x+c)(x+a)}{(c-b)(a-b)} + \frac{(x+a)(x+b)}{(a-c)(b-c)} = 1$ is an identify.

$x=-a \Rightarrow LHS = \frac{(b-a)(c-a)}{(b-a)(c-a)} + 0 + 0 = 1$

$x=-b$ & $x=-c$ also satisfy.

degree = 2

no of roots = 3 > (degree = 2)

→ (T)

2. If $\alpha^2 - 3\alpha - 5 = 0$ and $\beta^2 - 3\beta - 5 = 0$ then quadratic whose roots are α & β is $x^2 - 3x - 5 = 0$ & value of $\alpha\beta$ is $\frac{c}{a} = \frac{-5}{1}$,

also value of $\alpha^2\beta + \alpha\beta^2 + 7\alpha + 7\beta + 2$

$\alpha^2 - 3\alpha - 5 = 0$, $\beta^2 - 3\beta - 5 = 0$ → $x^2 - 3x - 5 = 0$ → α, β

$\alpha\beta(\alpha+\beta) + 7(\alpha+\beta) + 2$

$(-5)(3) + 7 \times 3 + 2 = -15 + 21 + 2 = 8$

$\alpha + \beta = -\frac{(-3)}{1} = 3$

Question Practice

Question

Find the value of m such that
$(m^2 - 3m + 2)x^2 + (m^2 - 5m + 4)x + (m^2 - 7m + 6) = 0$ is an identify.

A 1

B 2

C −2

D −1

Ans: A

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots are: (i) $\alpha^2 + 5$ & $\beta^2 + 5$

$x^2 - (\text{Sum})x + (\text{Prod.}) = 0$

$x^2 - 2x + 5 = 0$ ⟨ α, β

$\alpha + \beta = -\frac{(-2)}{1} = 2$

$\alpha\beta = \frac{5}{1} = 5$

SBS

$\alpha^2 + \beta^2 + 2(\alpha\beta) = 4$

$\alpha^2 + \beta^2 = 4 - 10 = -6$

$x^2 - (4)x + 20 = 0$

Sunita:

Sum $= \alpha^2 + 5 + \beta^2 + 5$

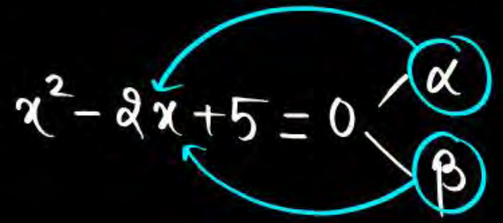
$= \alpha^2 + \beta^2 + 10 = -6 + 10 = 4$

Product $= (\alpha^2 + 5)(\beta^2 + 5)$

$= \alpha^2\beta^2 + 5\beta^2 + 5\alpha^2 + 25$

$= (\alpha\beta)^2 + 5(\beta^2 + \alpha^2) + 25$

$= 5^2 + 5(-6) + 25 = 20$



Sush:-

$\alpha^2+5,\ \beta^2+5 \Rightarrow 2\alpha, 2\beta$

$\alpha^2 - 2\alpha + 5 = 0$

$\alpha^2 + 5 = 2\alpha$

$\beta^2 + 5 = 2\beta$

$x^2 - (\text{Sum})x + (\text{Prod.}) = 0$

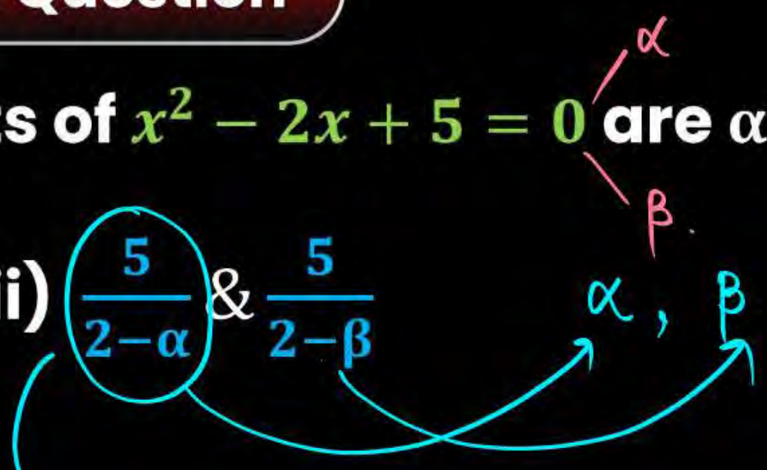
$x^2 - 2\left(\frac{\alpha+\beta}{2}\right)x + 4\left(\frac{\alpha\beta}{5}\right) = 0$

$x^2 - 4x + 20 = 0$

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots

are: (ii) $\frac{5}{2-\alpha}$ & $\frac{5}{2-\beta}$



Sush: $\alpha^2 - 2\alpha + 5 = 0$

$\alpha(\alpha-2) = -5$

$+\alpha(2-\alpha) = +5$

$$\alpha = \frac{5}{2-\alpha}$$

Ans: $x^2 - 2x + 5 = 0$

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots are: (iii) $\alpha^3 + \alpha^2 - \alpha + 22$ & $\beta^3 + 4\beta^2 - 7\beta + 35$

β, α → 5

$\alpha^2 - 2\alpha + 5 = 0$

$\alpha^2 = 2\alpha - 5$

$\alpha \times$ → $\alpha^3 = 2\alpha^2 - 5\alpha$

$= 2(2\alpha - 5) - 5\alpha$

$= 4\alpha - 10 - 5\alpha$

$\alpha^3 = -\alpha - 10$

$\alpha^3 + \alpha^2 - \alpha + 22 = 7$

$(-\beta - 10) + 4(2\beta - 5) - 7\beta + 35$

$-\beta - 10 + 8\beta - 20 - 7\beta + 35 = 5$ ✓

$-\alpha - 10 + 2\alpha - 5 - \alpha + 22 = 7$

$x^2 - (7+5)x + (7 \times 5) = 0$ → 7, 5

$x^2 - 12x + 35 = 0$

$\beta^2 - 2\beta + 5 = 0$

M-2

$\beta + 6$

$(\beta^2 - 2\beta + 5) \overline{\big) \beta^3 + 4\beta^2 - 7\beta + 35}$

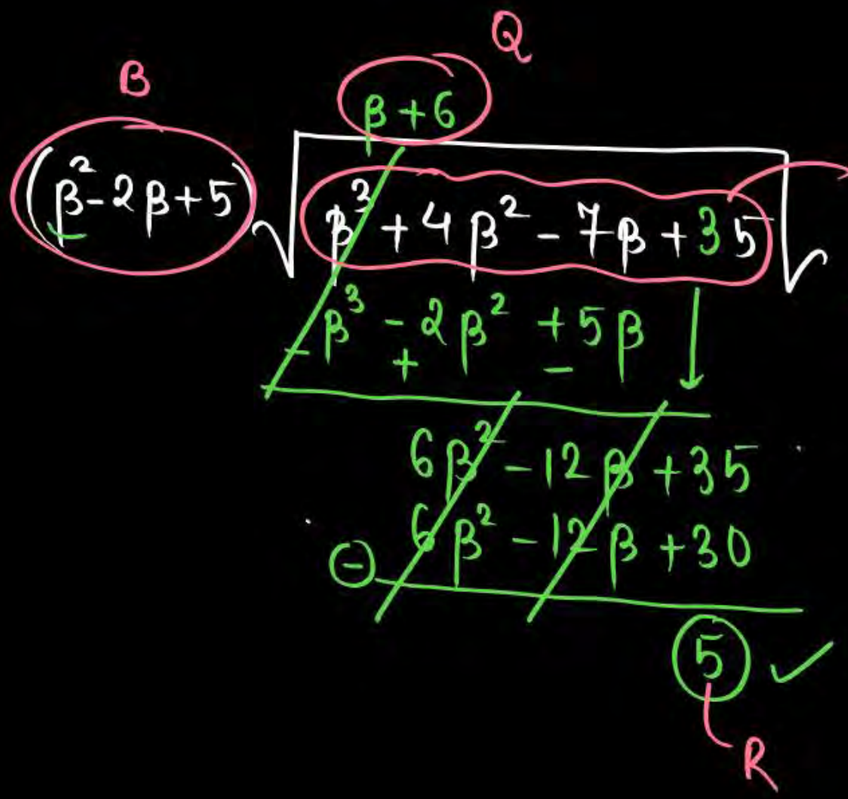
$\beta^3 - 2\beta^2 + 5\beta$

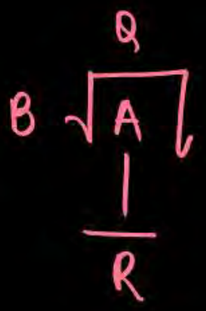
$6\beta^2 - 12\beta + 35$

$6\beta^2 - 12\beta + 30$

5 ✓

$\beta^2 - 2\beta + 5 = 0$





$\Downarrow$

$A = BQ + R$

$\beta^3+4\beta^2-7\beta+35 = (\beta^2-2\beta+5)(\beta+6) + 5$

0

$= 5$

Question [JEE (Adv.)-2015, Paper-2]

$\alpha \neq 0$

Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is (are) a subset(s) of S?

$D > 0$

$\alpha x^2 - x + \alpha = 0 \Rightarrow x_1, x_2$

$|x_1 - x_2|$

$$D.O.R = |x_1 - x_2| = \frac{\sqrt{D}}{|a|} = \frac{\sqrt{(-1)^2 - 4\alpha\alpha}}{|\alpha|} = \frac{\sqrt{1-4\alpha^2}}{|\alpha|} < 1$$

Solve.

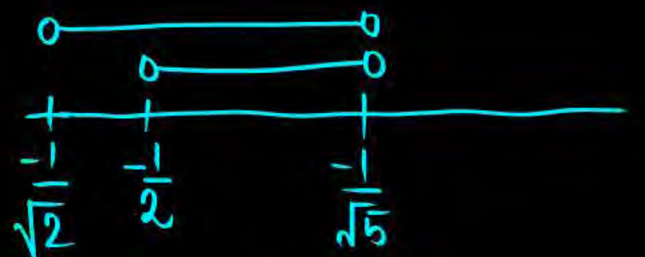
A $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{5}}\right)$

B $\left(-\frac{1}{\sqrt{5}}, 0\right)$

C $\left(0, \frac{1}{\sqrt{5}}\right)$

D $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

$$\alpha \in \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$$



irr + mod → ineq

$\frac{\sqrt{1-4\alpha^2}}{|\alpha|} < 1$ → Solve.

$|\alpha|$ → +ve

$\sqrt{1-4\alpha^2} < |\alpha|$

SBS

$1-4\alpha^2 < |\alpha|^2$

$1-4\alpha^2 < \alpha^2$

$0 < 5\alpha^2 - 1$

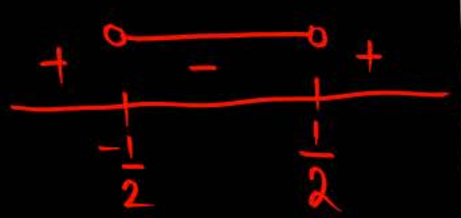
$0 < (\sqrt{5}\alpha - 1)(\sqrt{5}\alpha + 1)$

• domain

$1-4\alpha^2 > 0$

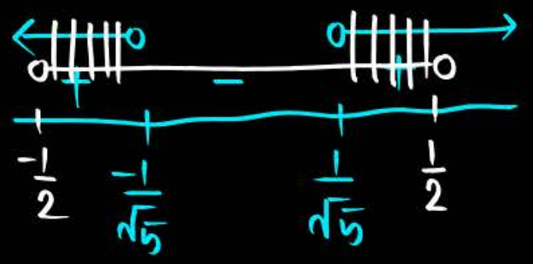
$4\alpha^2 - 1 < 0$

$(2\alpha-1)(2\alpha+1) < 0$



$|\alpha|^2 = |\alpha^2| = \alpha^2$

$5\alpha^2 - 1 = (\sqrt{5}\alpha)^2 - (1)^2$



$\alpha \in \left(-\frac{1}{2}, -\frac{1}{\sqrt{5}}\right) \cup \left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

DIBY-03

If α, β are the roots of $ax^2 + bx + c = 0$, then the equation whose roots are $2 + \alpha$ and $2 + \beta$ is

(A) $ax^2 + (b - 4a)x + (4a - 2b + c) = 0$ (B) $ax^2 - (b - 4a)x + (4a - 2b + c) = 0$

(C) $ax^2 + (b - 4a)x - (4a - 2b + c) = 0$ (D) $ax^2 - (b - 4a)x - (4a - 2b + c) = 0$

[Ans. : (A)]

$ax^2 + bx + c = 0$ (α, β)

$\alpha + \beta = -\frac{b}{a}$

$\alpha\beta = \frac{c}{a}$

$x^2 - (2+\alpha+2+\beta)x + (2+\alpha)(2+\beta) = 0$

$x^2 - (4 + \alpha + \beta)x + 4 + 2(\alpha+\beta) + \alpha\beta = 0$ ($2+\alpha$, $2+\beta$)

$x^2 - \left(4 - \frac{b}{a}\right)x + 4 + 2\left(\frac{-b}{a}\right) + \frac{c}{a} = 0$

$ax^2 + (b - 4a)x + 4a - 2b + c = 0$

DIBY-05

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$ (B) $\frac{2ac-b^2}{ac}$ (C) $\frac{b^2-2ac}{a^2c^2}$ (D) $\frac{b^2}{2}$

[Ans. : (C)]

$ax^2+bx+c=0$ — roots α, β

$\alpha+\beta = -\frac{b}{a}$

$\alpha\beta = \frac{c}{a}$

SBS

$\alpha^2+\beta^2+2\alpha\beta = \frac{b^2}{a^2}$

$\alpha^2+\beta^2 = \frac{b^2}{a^2} - \frac{2c}{a}$

$a\alpha^2+b\alpha+c=0$

$\alpha(a\alpha+b) = -c$

$a\alpha+b = -\frac{c}{\alpha}$

$-\frac{\alpha}{c} = \frac{1}{a\alpha+b}$

$\frac{\alpha^2}{c^2} = \frac{1}{(a\alpha+b)^2}$

$\frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2}$

$\frac{(\alpha^2+\beta^2)}{c^2} = \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{c^2}$

$= \frac{(b^2-2ac)}{a^2c^2}$

Question [IIT-JEE-2010, Paper-1]

Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

$\frac{\alpha}{\beta} \to \frac{1}{2}$, $\frac{\beta}{\alpha} \to 2$

(A) $(p^3+q)x^2 - (p^3+2q)x + (p^3+q) = 0$

$\to -18x^2 + 9x - 18 = 0 \to 2x^2 - x + 2 = 0$.

(B) $(p^3+q)x^2 - (p^3-2q)x + (p^3+q) = 0$

$\to -18x^2 + 45x - 18 = 0 \to 2x^2 - 5x + 2 = 0 \to \frac{1}{2}, 2$.

(C) $(p^3-q)x^2 - (5p^3-2q)x + (p^3-q) = 0$

(D) $(p^3-q)x^2 - (5p^3+2q)x + (p^3-q) = 0$

youtube method

$\alpha = 1, \beta = 2$

$3 = -p, \quad 1^3 + 2^3 = q$.

$p = -3 \quad q = 9$

$p^3 = -27$
$q = 9$
$\Big\} \; p^3 + q = -18$

Question [IIT-JEE-2010, Paper-1]

Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is

$$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)x + \left(\frac{\alpha}{\beta} \times \frac{\beta}{\alpha}\right) = 0$$

$$x^2 - \left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right)x + 1 = 0$$

$\frac{\alpha}{\beta}$, $\frac{\beta}{\alpha}$

$$\frac{(p^3-2q)}{3p(p^3+q)} = \left(\frac{p^3-2q}{p^3+q}\right)$$

SBS

$$\alpha+\beta = -p$$

$$\alpha^2+\beta^2+2\alpha\beta = p^2$$

$$\alpha^2+\beta^2 = p^2-2\alpha\beta$$

$$\alpha^2+\beta^2 = p^2 - 2\left(\frac{p^3+q}{3p}\right)$$

$$= \frac{3p^3-2p^3-2q}{3p}$$

$$= (p^3-2q)/3p$$

$$\alpha^3+\beta^3 = q$$

$$(\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta) = q$$

$$-p\,(p^2-2\alpha\beta-\alpha\beta) = q$$

$$p^2 - 3\alpha\beta = \frac{-q}{p}$$

$$p^2 + \frac{q}{p} = 3\alpha\beta$$

$$\frac{p^3+q}{3p} = \alpha\beta$$

$$x^2 - \left(\frac{\alpha^2+\beta^2}{\alpha\beta}\right)x + 1 = 0$$

$$\frac{(p^3-2q)}{3p(p^3+q)} = \left(\frac{p^3-2q}{p^3+q}\right)$$

$$x^2 - \left(\frac{p^3-2q}{p^3+q}\right)x + 1 = 0$$

Question [IIT-JEE-2007, Paper-1]

Let α, β be the roots of the equation $x^2 - px + r = 0$ and $\frac{\alpha}{2}, 2\beta$ be the roots of the equation $x^2 - qx + r = 0$. Then the value of r is

(A) $\frac{2}{9}(p-q)(2q-p)$

(B) $\frac{2}{9}(q-p)(2p-q)$

(C) $\frac{2}{9}(q-2p)(2q-p)$

(D) $\frac{2}{9}(2p-q)(2q-p)$

Hw.



SYMMETRIC FUNCTION OF ROOTS



Note-01

If we interchange position of α & β and expression remains same, then expression is called symmetric function of roots.

Ex. : $\alpha^2+\beta^2$, $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$, $\frac{1}{\alpha-4}+\frac{1}{\beta-4}$, $\alpha^2\beta+\beta^2\alpha$ etc.

Note-02

To find value of symmetric expression of roots, no need to find α & β separately. Just find $\alpha+\beta$ and $\alpha\beta$ and enjoy!!

Bumper Question

If α & β are roots of equation $x^2 - 4x - 8 = 0$ then find value of following

(i) $\alpha^2 + \beta^2$

(ii) $\left(\frac{\alpha^2}{\beta}\right)^{1/3} + \left(\frac{\beta^2}{\alpha}\right)^{1/3}$

(iii) $\frac{1}{(\alpha-4)^2} + \frac{1}{(\beta-4)^2}$

(iv) $\alpha^4 + \beta^4$

(v) $\alpha^5 + \beta^5$

(vi) $|\alpha^5 - \beta^5|$ hw

(vii) $\frac{\beta}{\alpha^2-8} + \frac{\alpha}{\beta^2-8}$ hw.

(viii) $\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)}$ hw

$\frac{1}{\beta^2} + \frac{1}{\alpha^2} = \frac{\alpha^2+\beta^2}{\alpha^2\beta^2} = \frac{32}{64} = \frac{1}{2}$

$(\alpha\beta)^{\frac{1}{3}} = (-8)^{\frac{1}{3}} = -2$

$x^2 - 4x - 8 = 0$ → α, β

\# $\alpha+\beta = 4$

\# $\alpha\beta = -8$

1. $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$
$= 4^2 - 2(-8)$
$= 16 + 16$
$= 32$.

2. $\frac{\alpha^{\frac{2}{3}}}{\beta^{\frac{1}{3}}} + \frac{\beta^{\frac{2}{3}}}{\alpha^{\frac{1}{3}}} = \frac{\alpha^{\frac{2}{3}}\cdot\alpha^{\frac{1}{3}} + \beta^{\frac{2}{3}}\cdot\beta^{\frac{1}{3}}}{\alpha^{\frac{1}{3}}\beta^{\frac{1}{3}}} = \frac{\alpha^1+\beta^1}{(\alpha\beta)^{\frac{1}{3}}}$

$= \frac{4}{-2} = -2$

3. $\alpha^2 - 4\alpha - 8 = 0$
$\alpha(\alpha-4) = 8$
$\frac{\alpha}{8} = \frac{1}{\alpha-4}$

$\left(\frac{1}{(\alpha-4)^2} + \frac{1}{(\beta-4)^2}\right)$
$\frac{\alpha^2}{64} + \frac{\beta^2}{64} = \frac{\alpha^2+\beta^2}{64} = \frac{32}{64} = \frac{1}{2}$

$x^2 - 4x - 8 = 0$ → roots α, β

$\alpha + \beta = 4$ # $\alpha\beta = -8$

$\alpha^2 + \beta^2 = 32$

$\alpha^5 + \beta^5 =$??

(iv)

Sumit:

SBS

$\alpha^4 + \beta^4 + 2\alpha^2\beta^2 = (32)^2$

$\alpha^4 + \beta^4 = (32)^2 - 2(-8)^2$

$= 32 \times 32 - 2 \times 8 \times 8$

$= 64 \times 14.$

$= 896.$

$\alpha^4 + \beta^4 = (\alpha+\beta)(\alpha^3+\beta^3) - (\beta\alpha^3 + \alpha\beta^3)$

forceful factorisation

$x^2 - 4x - 8 = 0$ → roots α, β

$\alpha + \beta = 4$ # $\alpha\beta = -8$

$\alpha^2 + \beta^2 = 32$

$\alpha^3 + \beta^3 = (\alpha+\beta)(\alpha^2+\beta^2-\alpha\beta)$

$= 4 \times (32 - (-8))$

$= 4 \times 40$

$= 160$

$\alpha^5 + \beta^5 = (\alpha^2+\beta^2)(\alpha^3+\beta^3) - (\beta^2\alpha^3 + \alpha^2\beta^3)$

$= (\alpha^2+\beta^2)(\alpha^3+\beta^3) - \alpha^2\beta^2(\alpha+\beta)$

$\alpha^5 + \beta^5 = 32 \times 160 - (-8)^2(4)$

$\alpha^5 + \beta^5 = (\alpha+\beta)(\alpha^4+\beta^4) - (\beta\alpha^4 + \alpha\beta^4)$

$= (\alpha+\beta)(\alpha^4+\beta^4) - \alpha\beta(\alpha^3+\beta^3)$

$\alpha^7 + \beta^7 = (\alpha^3+\beta^3)(\alpha^4+\beta^4) - (\beta^3\alpha^4 + \alpha^3\beta^4)$

Question [JEE (Adv.-2020), Paper-1]

Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$.

Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$.

(A) 0

(B) 8000

(C) 8080

(D) 16000

Hw.
90% DIBY

DO IT BY YOURSELF (DIBY)

$\sqrt{10} = \sqrt{p^2-3p} \Leftarrow \sqrt{10} = \sqrt{p^2-4\times1\times\frac{3p}{4}} \Leftarrow \sqrt{10} = \frac{\sqrt{D}}{|a|}$ (SBS ✓, $|a| = 1$)

DIBY-04

If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set

(A) $\{2, -5\}$ (B) $\{-3, 2\}$ (C) $\{3, -5\}$ (D) $\{-2, 5\}$

[Ans. : (D)]

DIBY-05

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$ (B) $\frac{2ac-b^2}{ac}$ (C) $\frac{b^2-2ac}{a^2c^2}$ (D) $\frac{b^2}{2}$

[Ans. : (C)]

DIBY-06 (Roots in bracket)

Let α and β be the roots of the quadratic equation $(x-2)(x-3) + (x-3)(x+1) + (x+1)(x-2) = 0$. Find the value of

$$\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}$$

[Ans. : 0]

DIBY-07

If α, β are the roots of $ax^2 + bx + c = 0, (a \neq 0)$ and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0, (A \neq 0)$ for some constant δ, then prove that, $\frac{b^2-4ac}{a^2} = \frac{B^2-4AC}{A^2}$.

[JEE-2000]

DIBY-08

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

(A) 1 (B) 3 (C) 4 (D) 2

[Ans. : (D)]

DIBY-09

Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}(\alpha^6 + \beta^6)$, is: [JEE Mains-2024]

(A) $x^2 - 190x + 9466 = 0$ (B) $x^2 - 195x + 9466 = 0$

(C) $x^2 - 195x + 9506 = 0$ (D) $x^2 - 180x + 9506 = 0$

[Ans. : (C)]

DIBY-11

Let $\alpha \in \mathbb{R}$ and let α, β be the roots of the equation $x^2 + 60^{\frac{1}{4}}x + a = 0$.
If $\alpha^4 + \beta^4 = -30$, then the product of all possible values of a is [JEE Mains-2023]

[Ans. : 45]

DIBY-12

If 2 and 6 are the roots of the equation $ax^2 + bx + 1 = 0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is : [JEE Mains-2024]

(A) $x^2 + 8x + 12 = 0$
(B) $2x^2 + 11x + 12 = 0$
(C) $4x^2 + 14x + 12 = 0$
(D) $x^2 + 10x + 16 = 0$

[Ans. : (A)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Basic Maths (4 Sittings Each of 1 Hr.)

Exercise-03(BRIDGE): COMPLETE

Exercise-04(JEE ADV. LEVEL):COMPLETE

It is never about end result

It's always about Journey

Thank You

#futureIITians